

CSE 715: Computer Graphics

Chapter 5 — Two-Dimensional Viewing and Clipping

Complete Model Answers (2020–2024)

University of Chittagong — 7th Semester B.Sc. (Engg.) Examination

Source: Schaum's Outline of Computer Graphics (Xiang & Plastock), Chapter 5 — notation follows the book exactly (region-code bit order, w_x/w_y , v_x/v_y , $x_{\min}/x_{\max}/y_{\min}/y_{\max}$)

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Theoretical Framework — Reusable Across All Years

Chapter 5 shows up as **Q4 (Section A, every year)** for Cohen–Sutherland clipping and Sutherland–Hodgman/Weiler–Atherton polygon clipping, and inside **Q5 (Section B, every year)** for the Normalization (window-to-viewport) transformation. Convex/concave polygon identification is bundled wherever the paper happens to place it (Q4, Q6, or Q7 depending on the year) but is answered here since it is §5.4 material.

Window-to-Viewport (Normalization) Transformation — §5.1

A **window** is specified by four world coordinates $wx_{\min}, wx_{\max}, wy_{\min}, wy_{\max}$. A **viewport** is specified by four normalized device coordinates $vx_{\min}, vx_{\max}, vy_{\min}, vy_{\max}$. To map $(wx, wy) \rightarrow (vx, vy)$ preserving relative placement:

$$vx = \frac{vx_{\max} - vx_{\min}}{wx_{\max} - wx_{\min}}(wx - wx_{\min}) + vx_{\min}, \quad vy = \frac{vy_{\max} - vy_{\min}}{wy_{\max} - wy_{\min}}(wy - wy_{\min}) + vy_{\min}$$

As a translate–scale–translate composite matrix N with $s_x = \frac{vx_{\max} - vx_{\min}}{wx_{\max} - wx_{\min}}$, $s_y = \frac{vy_{\max} - vy_{\min}}{wy_{\max} - wy_{\min}}$:

$$N = \begin{pmatrix} 1 & 0 & vx_{\min} \\ 0 & 1 & vy_{\min} \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} s_x & 0 & 0 \\ 0 & s_y & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & -wx_{\min} \\ 0 & 1 & -wy_{\min} \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} s_x & 0 & -s_x wx_{\min} + vx_{\min} \\ 0 & s_y & -s_y wy_{\min} + vy_{\min} \\ 0 & 0 & 1 \end{pmatrix}$$

Workstation transformation is the same operation a second time: NDC $\rightarrow$ physical device pixels. **Preserving aspect ratio** (only a window given, no viewport): compute window aspect $a_w = \frac{\text{width}}{\text{height}}$. If $a_w \geq 1$ (wide), use the full x -extent $[0, 1]$ and y -extent $[0, 1/a_w]$. If $a_w < 1$ (tall), use the full y -extent $[0, 1]$ and x -extent $[0, a_w]$. Either way $s_x = s_y$ (uniform scale $\Rightarrow$ no distortion).

Point and Line Clipping — Cohen–Sutherland Algorithm (§5.2–5.3)

Point clipping: (x, y) is inside the window iff $x_{\min} \leq x \leq x_{\max}$ and $y_{\min} \leq y \leq y_{\max}$.

4-bit region code (exact book order — **TOP, BOTTOM, RIGHT, LEFT**), using $\text{sign}(a) = 1$ if $a > 0$, else 0:

Bit 1 (TOP) = $\text{sign}(y - y_{\max})$ Bit 2 (BOTTOM) = $\text{sign}(y_{\min} - y)$ Bit 3 (RIGHT) = $\text{sign}(x - x_{\max})$

A point with code 0000 is inside the window.

Three clipping categories for a line from (x_1, y_1) to (x_2, y_2) :

1. **Visible** — both endpoint codes are 0000.
2. **Not visible** — bitwise AND of the two codes $\neq$ 0000 (both endpoints lie outside the *same* boundary).
3. **Clipping candidate** — neither of the above (AND = 0000 but not both codes 0000).

Algorithm (iterative): pick an endpoint with a nonzero code. Push each set bit across its boundary in turn: bit TOP $\rightarrow$ intersect with $y = y_{\max}$; bit BOTTOM $\rightarrow y = y_{\min}$; bit RIGHT $\rightarrow x = x_{\max}$; bit LEFT $\rightarrow x = x_{\min}$. Replace that endpoint with the intersection point, recompute its code, and re-classify. Repeat until the segment is Visible or Not Visible.

Intersection with vertical boundary $x = x_i : y_i = y_1 + m(x_i - x_1)$ with horizontal $y = y_i : x_i = x_1 + (y_i - y_1)/m$

Polygon Clipping — Convexity, Orientation, Left/Right Test (§5.4)

Convex polygon: the segment joining any two interior points lies entirely inside the polygon. A **concave** polygon is any polygon that is not convex (at least one interior segment exits the boundary — typically visible as a “notch” or reflex vertex $> 180^\circ$).

Positive orientation: a polygon $P_1, \dots, P_N$ is positively oriented if traversing the vertices in order produces a **counterclockwise** circuit — equivalently, the left hand of someone walking along any directed edge $\overrightarrow{P_{i-1}P_i}$ points *inside* the polygon.

Left/right-of-line test: for a directed edge $A(x_1, y_1) \rightarrow B(x_2, y_2)$ and a point $P(x, y)$:

$$C = (x_2 - x_1)(y - y_1) - (y_2 - y_1)(x - x_1)$$

$C > 0 \Rightarrow P$ is to the **left** of $\overrightarrow{AB}$; $C < 0 \Rightarrow P$ is to the **right**. For a positively-oriented convex clipping polygon: P outside if to the right of *any* edge, inside if to the left of *every* edge.

Sutherland–Hodgman Algorithm (§5.4)

Clips a (possibly concave) subject polygon against one edge E (endpoints A, B) of a positively-oriented *convex* clipping polygon at a time, re-feeding the output into the next edge. For each subject edge $\overrightarrow{P_{i-1}P_i}$, using the left/right test above against E :

1. Both P_{i-1}, P_i **left** of $E \Rightarrow$ output P_i .
2. Both **right** of $E \Rightarrow$ output nothing.
3. P_{i-1} left, P_i right (**exiting**) $\Rightarrow$ output the intersection point I of $\overrightarrow{P_{i-1}P_i}$ with (the extended) edge E .
4. P_{i-1} right, P_i left (**entering**) $\Rightarrow$ output I , then P_i .

Process all edges of the subject polygon (closing back to P_1), then repeat against the clip polygon’s next edge. **Caveat:** the algorithm can join disconnected output pieces with spurious double-back edges when the subject polygon splits into multiple parts

— these coincide in opposite directions and can be removed in a post-processing pass, or avoided entirely by using Weiler–Atherton.

Weiler–Atherton Algorithm (§5.4)

Handles concave clip windows and correctly outputs multiple disjoint sub-polygons (Sutherland–Hodgman’s weakness). Call the clip window the **clip polygon** and the polygon being clipped the **subject polygon**.

1. Start at any subject-polygon vertex; trace its boundary.
2. If an edge **enters** the clip polygon, record the intersection point and continue tracing the subject polygon.
3. If an edge **leaves** the clip polygon, record the intersection point, then make a **right turn** onto the clip polygon boundary (swap roles: now trace what was the clip polygon, treating it as the subject).
4. Whenever the trace closes into a loop, output that closed loop as one result sub-polygon; resume from the next not-yet-traced recorded intersection.
5. Terminate when the entire original subject-polygon border has been traced exactly once.

Examination 2024 — Q5(a,b,c) & Q6(a) (Section B)

Question breakdown:

- Q5(a) Find the normalization transformation that maps a window with lower-left corner $(1, 1)$ and upper-right corner $(3, 5)$ onto: (i) a viewport that is the entire NDC screen; (ii) a viewport with lower-left corner $(0, 0)$ and upper-right corner (v_x, v_y) . (3)
- Q5(b) Clipping window $x_{\min} = 10, x_{\max} = 30, y_{\min} = 10, y_{\max} = 25$. Compute the 4-bit Cohen–Sutherland region code (TOP, BOTTOM, RIGHT, LEFT) for $P_1(5, 5), P_2(20, 30), P_3(35, 15), P_4(15, 8), P_5(12, 20)$. Then determine whether $A(5, 5)–B(12, 20)$ is visible, not visible, or requires clipping. (3)
- Q5(c) *(Related — Circular Clipping, not core §5 topic but bundled in this question)* Circular window center $C(10, 10)$, radius $r = 6$; points $p_1(2, 8), p_2(8, 16)$. Determine if the segment is completely inside, completely outside, or intersecting; if intersecting, compute the intersection point(s) and the clipped visible portion. (3)
- Q6(a) Clipping window: convex quadrilateral $A(1, 1), B(5, 1), C(5, 4), D(1, 4)$. Subject polygon: triangle $P(2, 2), Q(4, 2), R(3, 5)$. Perform Weiler–Atherton polygon clipping: (i) show entry/exit points; (ii) list clipped-polygon vertices in CCW order. (3)

Q5(a) — Normalization, Window (1, 1)–(3, 5) [3 marks]

$wx_{\min} = 1, wx_{\max} = 3, wy_{\min} = 1, wy_{\max} = 5$ (width 2, height 4).

(i) **Viewport = entire NDC** ($vx_{\min} = 0, vx_{\max} = 1, vy_{\min} = 0, vy_{\max} = 1$): $s_x = \frac{1}{2}, s_y = \frac{1}{4}$.

$$N = \begin{pmatrix} \frac{1}{2} & 0 & -\frac{1}{2} \\ 0 & \frac{1}{4} & -\frac{1}{4} \\ 0 & 0 & 1 \end{pmatrix}$$

(ii) **Viewport** $(0, 0)$ – (v_x, v_y) (symbolic): $s_x = \frac{v_x}{2}, s_y = \frac{v_y}{4}$.

$$N = \begin{pmatrix} \frac{v_x}{2} & 0 & -\frac{v_x}{2} \\ 0 & \frac{v_y}{4} & -\frac{v_y}{4} \\ 0 & 0 & 1 \end{pmatrix}$$

Q5(b) — Region Codes + Visibility Classification [3 marks]

Window: $x_{\min} = 10, x_{\max} = 30, y_{\min} = 10, y_{\max} = 25$. Bit order (TOP,BOTTOM,RIGHT,LEFT):

Point	x, y	TOP	BOTTOM	RIGHT	LEFT	Code
P_1	(5, 5)	0	1	0	1	0101
P_2	(20, 30)	1	0	0	0	1000
P_3	(35, 15)	0	0	1	0	0010
P_4	(15, 8)	0	1	0	0	0100
P_5	(12, 20)	0	0	0	0	0000 (inside)

$A(5, 5) = P_1, B(12, 20) = P_5$: codes 0101 AND 0000 = 0000, and not both 0000 $\Rightarrow$ **requires clipping** (Category 3). A is outside on both LEFT and BOTTOM; clip against $x = x_{\min} = 10$ first: direction $B - A = (7, 15)$, $t = \frac{10-5}{7} = \frac{5}{7}$, $y = 5 + 15 \cdot \frac{5}{7} = \frac{110}{7} \approx 15.71$. Intersection $I_1 = (10, \frac{110}{7})$, code 0000 (inside). Segment I_1 – B : both 0000 $\Rightarrow$ visible. **Visible portion:** $(10, 15.71)$ to $(12, 20)$.

Q5(c) — Circular Clipping Window [3 marks, Related]

Circle: $(x - 10)^2 + (y - 10)^2 = 36$. Check endpoints: $p_1(2, 8)$: $(2 - 10)^2 + (8 - 10)^2 = 64 + 4 = 68 > 36$ (outside). $p_2(8, 16)$: $(8 - 10)^2 + (16 - 10)^2 = 4 + 36 = 40 > 36$ (outside). Both endpoints are outside, but the chord may still pass through the circle — parametrize $p_1 \rightarrow p_2$: $x = 2 + 6t, y = 8 + 8t, t \in [0, 1]$. Substituting:

$$(6t-8)^2 + (8t-2)^2 = 36 \Rightarrow 100t^2 - 128t + 32 = 0 \Rightarrow 25t^2 - 32t + 8 = 0 \Rightarrow t = \frac{16 \pm 2\sqrt{14}}{25}$$

$t_1 \approx 0.341, t_2 \approx 0.939$ — both in $[0, 1]$, so the segment **intersects** the circle at two

points (enters then exits):

$$I_1 \approx (4.04, 10.73) \quad I_2 \approx (7.64, 15.51)$$

The clipped (visible, inside-circle) portion is the chord from I_1 to I_2 ; the two end stubs p_1-I_1 and I_2-p_2 lie outside the circle and are discarded.

Q6(a) — Weiler–Atherton: Triangle vs. Rectangle [3 marks]

Clip window $ABCD$: $x \in [1, 5]$, $y \in [1, 4]$. Subject triangle $P(2, 2), Q(4, 2), R(3, 5)$: P, Q inside; R outside (above $y_{\max} = 4$). **Exit point** (edge $Q \rightarrow R$ leaves through top $y = 4$): direction $R - Q = (-1, 3)$, $t = \frac{4-2}{3} = \frac{2}{3}$, $x = 4 - \frac{2}{3} = \frac{10}{3}$. $E_1 = (\frac{10}{3}, 4)$. **Entry point** (edge $R \rightarrow P$ re-enters through top $y = 4$): direction $P - R = (-1, -3)$, $t = \frac{4-5}{-3} = \frac{1}{3}$, $x = 3 - \frac{1}{3} = \frac{8}{3}$. $E_2 = (\frac{8}{3}, 4)$. Trace: $P(2, 2) \rightarrow Q(4, 2)$ [inside] $\rightarrow$ exit at $E_1 \rightarrow$ right turn onto window's top edge $\rightarrow$ entry at $E_2 \rightarrow$ resume subject trace back to P . **Clipped polygon (CCW)**: $P(2, 2), Q(4, 2), E_1(\frac{10}{3}, 4), E_2(\frac{8}{3}, 4)$ — a quadrilateral (verified positively-oriented via the shoelace formula).

Examination 2023 — Q4(a) (Section A) & Q8(a) (Section B)

Question breakdown:

- Q4(a) Polygon $A(1, 1), B(4, 1), C(4, 4), D(1, 4)$ (CCW). Clipping window: rectangle $W(2, 2), X(3, 2), Y(3, 3), Z(2, 3)$. Using Sutherland–Hodgman, calculate the clipped-polygon vertices step by step for each edge of the clipping window.
- Q8(a) Perform the Cohen–Sutherland algorithm for line segment endpoints $(5, 5)$ and $(15, 10)$ in a clipping window with boundaries $x_{\min} = 0, y_{\min} = 0, x_{\max} = 10, y_{\max} = 10$. (3.5)

Q4(a) — Sutherland–Hodgman: 3×3 Square vs. 1×1 Window

Subject: $A(1, 1), B(4, 1), C(4, 4), D(1, 4)$. Clip window edges (each processed in turn against the whole running polygon): **Clip against left edge** $x = 2$ (keep $x \geq 2$): $A \rightarrow B$ exits/enters at $(2, 1)$, keep $B(4, 1)$; $B \rightarrow C$ both in, keep $C(4, 4)$; $C \rightarrow D$ exit at $(2, 4)$; $D \rightarrow A$ both out. Result: $(2, 1), (4, 1), (4, 4), (2, 4)$. **Clip against bottom edge** $y = 2$ (keep $y \geq 2$): result $(4, 2), (4, 4), (2, 4), (2, 2)$. **Clip against right edge** $x = 3$ (keep $x \leq 3$): result $(3, 4), (2, 4), (2, 2), (3, 2)$. **Clip against top edge** $y = 3$ (keep $y \leq 3$): result $(2, 3), (2, 2), (3, 2), (3, 3)$. **Final clipped polygon**: $(2, 3), (2, 2), (3, 2), (3, 3)$ — exactly the window Z, W, X, Y (cyclically shifted), confirming the correct result since the 1×1 window lies entirely inside the 3×3 square.

Q8(a) — Cohen–Sutherland: (5, 5)–(15, 10), Window (0, 0)–(10, 10) [3.5 marks]

$P_1(5, 5)$: code 0000 (inside). $P_2(15, 10)$: $x = 15 > 10 \Rightarrow \text{RIGHT} = 1$; $y = 10$ not $> 10 \Rightarrow \text{TOP} = 0$. Code 0010. AND = 0000, one endpoint inside $\Rightarrow$ clip P_2 against $x = x_{\max} = 10$: direction (10, 5), $t = \frac{10-5}{10} = 0.5$, $y = 5 + 5(0.5) = 7.5$. $I = (10, 7.5)$, code 0000. **Visible segment: (5, 5) to (10, 7.5).**

Examination 2022 — Q4(a,b,c) (Section A) & Q7(b) (Section B)

Question breakdown:

- Q4(a) Find region codes for $A(6, 2)$, $B(3, 8)$, $C(-1, 2)$, $D(2, 4)$; window $x_{\min} = 2$, $x_{\max} = 5$, $y_{\min} = 1$, $y_{\max} = 9$. Clip AB and CD using Cohen–Sutherland. (3)
- Q4(b) Using Sutherland–Hodgman, show the clipping steps for the given polygon/window figure (P_1 – P_8 vs. shaded clipping window). (4)
- Q4(c) Find the normalization transformation from a window (lower-left/upper-right corners as given in the paper) onto the NDC screen so that aspect ratios are preserved. (2)
- Q7(b) Draw the example diagram of the convex and concave polygon. (1)

Q4(a) — Region Codes + Clip AB , CD [3 marks]

Window: $x_{\min} = 2$, $x_{\max} = 5$, $y_{\min} = 1$, $y_{\max} = 9$.

Point	TOP	BOTTOM	RIGHT	LEFT	Code
$A(6, 2)$	0	0	1	0	0010
$B(3, 8)$	0	0	0	0	0000 (inside)
$C(-1, 2)$	0	0	0	1	0001
$D(2, 4)$	0	0	0	0	0000 (inside)

Clip AB : A outside (RIGHT), clip against $x = x_{\max} = 5$: direction $B - A = (-3, 6)$, $t = \frac{5-6}{-3} = \frac{1}{3}$, $y = 2 + 6 \cdot \frac{1}{3} = 4$. $I = (5, 4)$, code 0000. Segment I – B : both 0000 $\Rightarrow$ visible. **Clipped $AB = (5, 4)$ to $(3, 8)$.** **Clip CD :** C outside (LEFT), clip against $x = x_{\min} = 2$: direction $D - C = (3, 2)$, $t = \frac{2-(-1)}{3} = 1$ — this reaches D exactly, i.e. the line touches the boundary precisely at $D(2, 4)$. **The entire segment CD except the single point D is clipped away** (the line only grazes the window at its corner-boundary $x = 2$).

Q4(b) — Sutherland–Hodgman on Figure P_1 – P_8 [4 marks]

No numeric coordinates are given — the figure shows an 8-vertex subject polygon $P_1, \dots, P_8$ partly overlapping a rectangular shaded clipping window. **Method (apply directly to your own paper's figure):** process one clip-window edge at a time (left, right, top, bottom in any consistent order). For each edge E , walk the subject polygon's edges $\overline{P_{i-1}P_i}$ using the left/right test (§1.3): both endpoints left of $E \Rightarrow$ output P_i ; both right $\Rightarrow$ output nothing; left-to-right (exiting) $\Rightarrow$ output the intersection point; right-to-left (entering) $\Rightarrow$ output the intersection point then P_i . Re-feed the output vertex list into the next clip edge. From the figure, P_1, P_2, P_5, P_6 sit outside the window while P_3, P_4, P_7, P_8 sit on/inside it — so the edges touching the window boundary ($P_2-P_3, P_4-P_5, P_6-P_7, P_8-P_1$) are exactly where new intersection vertices get generated at each clipping stage.

Q4(c) — Normalization, Aspect Ratio Preserved [2 marks]

Scan note: the printed corners are partly illegible; read here as lower-left (1, 1), upper-right (3, 4) (width 2, height 3) — the only reading consistent with a valid (non-degenerate) rectangle and with "aspect ratio preserved" phrasing (matches the book's own Solved Problem 5.4 pattern). If your copy shows different corners, the method below is unchanged — just substitute your numbers.

$wx_{\min} = 1, wx_{\max} = 3, wy_{\min} = 1, wy_{\max} = 4$: width 2, height 3, $a_w = \frac{2}{3} < 1$ (taller than wide) $\Rightarrow$ use full y -extent $[0, 1]$, x -extent $[0, \frac{2}{3}]$. $s_x = \frac{2/3}{2} = \frac{1}{3}$, $s_y = \frac{1}{3} = \frac{1}{3}$ (equal — aspect preserved).

$$N = \begin{pmatrix} \frac{1}{3} & 0 & -\frac{1}{3} \\ 0 & \frac{1}{3} & -\frac{1}{3} \\ 0 & 0 & 1 \end{pmatrix}$$

Q7(b) — Convex vs. Concave Diagram [1 mark]

Convex: every interior-point-to-interior-point segment stays inside (e.g. a triangle, rectangle, regular pentagon). **Concave:** at least one such segment exits the boundary, visible as a notch/reflex vertex ($> 180^\circ$ interior angle), e.g. an arrow or star shape. See §1.3 definition and Fig. 5-7 (Schaum's).

Examination 2021 — Q4(a,b,c) (Section A) & Q7(d) (Section B)

Question breakdown:

Q4(a) Region codes for $M(8, 11), N(16, 7), P(6, 8), Q(3, 5)$; window from rectangle $A(4, 4), B(14, 4), C(14, 10), D(4, 10)$

Clip MN and PQ using Cohen–Sutherland.

- Q4(b) Polygon $abcdefghij\overline{kl}$ vs. viewing window $MNOP$ (figure). Using Sutherland–Hodgman, clip in order: (a) left edge PM , (b) bottom edge PO . (3.25)
- Q4(c) Identify the convex and concave polygons from three given shapes; briefly explain your selection. (1.5)
- Q7(d) Find the normalization transformation that maps a window with lower-left $(1, 1)$ and upper-right $(2, 2)$ onto a viewport with lower-left $(0, 0)$ and upper-right $(1/2, 1/2)$. (4)

Q4(a) — Region Codes + Clip MN , PQ

Window (from $A(4, 4), B(14, 4), C(14, 10), D(4, 10)$): $x_{\min} = 4, x_{\max} = 14, y_{\min} = 4, y_{\max} = 10$.

Point	TOP	BOTTOM	RIGHT	LEFT	Code
$M(8, 11)$	1	0	0	0	1000
$N(16, 7)$	0	0	1	0	0010
$P(6, 8)$	0	0	0	0	0000 (inside)
$Q(3, 5)$	0	0	0	1	0001

Clip MN : M outside (TOP), clip against $y = y_{\max} = 10$: direction $N - M = (8, -4)$, $t = \frac{10-11}{-4} = 0.25$, $x = 8 + 8(0.25) = 10$. $I_1 = (10, 10)$, code 0000. Segment $I_1 - N$: N still RIGHT-coded, clip against $x = x_{\max} = 14$: direction $(6, -3)$, $t = \frac{14-10}{6} = \frac{2}{3}$, $y = 10 - 3 \cdot \frac{2}{3} = 8$. $I_2 = (14, 8)$, code 0000. **Clipped MN = $(10, 10)$ to $(14, 8)$.**

Clip PQ : Q outside (LEFT), clip against $x = x_{\min} = 4$: direction $P - Q = (3, 3)$, $t = \frac{4-3}{3} = \frac{1}{3}$, $y = 5 + 3 \cdot \frac{1}{3} = 6$. $I = (4, 6)$, code 0000. **Clipped PQ = $(4, 6)$ to $(6, 8)$.**

Q4(b) — Sutherland–Hodgman on Figure $abcdefghij\overline{kl}$ vs. $MNOP$ [3.25 marks]

Figure-only question (no numeric coordinates) — apply the method of §1.4/§1.5 directly to the printed figure, processing the two named edges in the required order:

- Left clip edge $\overline{PM}$:** for each subject edge, test both endpoints' left/right status against $\overline{PM}$ (the window's left boundary). Vertices to the left survive as-is; edges crossing the boundary generate a new intersection vertex; edges entirely to the right (outside) are dropped.
- Bottom clip edge $\overline{PO}$:** re-feed the output list from step (i) and repeat the same left/right test against $\overline{PO}$ (the window's bottom boundary).

The polygon's re-entrant "V" notch (visible between the labeled vertices in the middle of the figure) means several edges cross each clip boundary twice — track each subject edge independently rather than assuming one crossing per edge.

Q4(c) — Convex/Concave Identification [1.5 marks]

Three shapes given: (a) a trapezoid — **convex** (no notches, every interior segment stays inside); (b) a zig-zag/arrow-like polygon — **concave** (has a reflex vertex $> 180^\circ$, an interior segment across the notch exits the boundary); (c) a rounded hexagon-like shape — **convex** (all interior angles $< 180^\circ$). Justify each using the definition in §1.3: check every vertex's interior angle, or equivalently check whether any two interior points' connecting segment leaves the shape.

Q7(d) — Normalization, Window $(1, 1)$ – $(2, 2) \rightarrow$ Viewport $(0, 0)$ – $(1/2, 1/2)$ [4 marks]

$wx_{\min} = 1, wx_{\max} = 2, wy_{\min} = 1, wy_{\max} = 2$ (unit window); $vx_{\min} = 0, vx_{\max} = \frac{1}{2}, vy_{\min} = 0, vy_{\max} = \frac{1}{2}$. $s_x = \frac{1/2}{1} = \frac{1}{2}, s_y = \frac{1}{2}$.

$$N = \begin{pmatrix} \frac{1}{2} & 0 & -\frac{1}{2} \\ 0 & \frac{1}{2} & -\frac{1}{2} \\ 0 & 0 & 1 \end{pmatrix}$$

Examination 2020 — Q5(a,b,c) & Q6(b,c) (Group-B)**Question breakdown:**

- Q5(a) Find the workstation transformation that maps the NDC screen onto a physical device with x extent 0–199 and y extent 0–639, origin at the lower-left corner. (2)
- Q5(b) Find the normalization transformation mapping window $(1, 1)$ – $(2, 2)$ onto viewport $(0, 0)$ – $(1/2, 1/2)$. (2.75)
- Q5(c) Briefly explain the Cohen–Sutherland algorithm. (1)
- Q6(b) Region codes for $A(6, 2), B(3, 8), C(-1, 2), D(2, 4)$; window $x_{\min} = 1, x_{\max} = 5, y_{\min} = 1, y_{\max} = 7$. Clip AB, CD . (4)
- Q6(c) Identify the convex and concave polygons from four given figures; justify your selection. (2)

Q5(a) — Workstation Transformation [2 marks]

Source is the entire NDC unit square ($wx_{\min} = 0, wx_{\max} = 1, wy_{\min} = 0, wy_{\max} = 1$); target device $vx_{\min} = 0, vx_{\max} = 199, vy_{\min} = 0, vy_{\max} = 639$, origin at the lower-left

(so no axis flip needed). $s_x = 199$, $s_y = 639$.

$$N = \begin{pmatrix} 199 & 0 & 0 \\ 0 & 639 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Q5(b) — Normalization, Window $(1, 1)$ – $(2, 2) \rightarrow$ Viewport $(0, 0)$ – $(1/2, 1/2)$ [2.75 marks]

Identical setup and result to 2021 Q7(d) above:

$$N = \begin{pmatrix} \frac{1}{2} & 0 & -\frac{1}{2} \\ 0 & \frac{1}{2} & -\frac{1}{2} \\ 0 & 0 & 1 \end{pmatrix}$$

Q5(c) — Briefly Explain Cohen–Sutherland [1 mark]

Assign each line endpoint a 4-bit region code (TOP/BOTTOM/RIGHT/LEFT) based on its position relative to the clipping window. If both codes are 0000 the line is visible; if the bitwise AND of the two codes is nonzero the line is entirely outside (not visible); otherwise, repeatedly replace an outside endpoint with its intersection point against the boundary indicated by its set bit(s), recomputing the code each time, until the segment resolves to visible or not-visible.

Q6(b) — Region Codes + Clip AB , CD [4 marks]

Window: $x_{\min} = 1, x_{\max} = 5, y_{\min} = 1, y_{\max} = 7$.

Point	TOP	BOTTOM	RIGHT	LEFT	Code
$A(6, 2)$	0	0	1	0	0010
$B(3, 8)$	1	0	0	0	1000
$C(-1, 2)$	0	0	0	1	0001
$D(2, 4)$	0	0	0	0	0000 (inside)

Clip AB : A outside (RIGHT), clip against $x = x_{\max} = 5$: direction $B - A = (-3, 6)$, $t = \frac{5-6}{-3} = \frac{1}{3}$, $y = 2 + 6 \cdot \frac{1}{3} = 4$. $I_1 = (5, 4)$, code 0000. Segment I_1 – B : B still TOP-coded, clip against $y = y_{\max} = 7$: direction $(-2, 4)$, $t = \frac{7-4}{4} = 0.75$, $x = 5 - 2(0.75) = 3.5$. $I_2 = (3.5, 7)$, code 0000. **Clipped $AB = (5, 4)$ to $(3.5, 7)$.** **Clip CD :** C outside (LEFT), clip against $x = x_{\min} = 1$: direction $D - C = (3, 2)$, $t = \frac{1-(-1)}{3} = \frac{2}{3}$, $y = 2 + 2 \cdot \frac{2}{3} = \frac{10}{3}$. $I = (1, \frac{10}{3})$, code 0000. **Clipped $CD = (1, 3.33)$ to $(2, 4)$.**

Q6(c) — Convex/Concave Identification [2 marks]

Four shapes: (i) a trapezoid — **convex**; (ii) a rounded/dented hexagon-like shape — **convex** if all vertex angles $< 180^\circ$, else concave (check the specific figure’s dent); (iii) a zig-zag arrow shape — **concave** (reflex vertex present); (iv) a regular pentagon — **convex**. Justification method: for each vertex, check whether the interior angle exceeds 180° (concave) or not (convex); equivalently, test whether any interior-to-interior segment exits the polygon (§1.3).

Quick Summary — Exam Patterns for Chapter 5

Pattern	Years	Method to memorise
Cohen–Sutherland region codes (TOP,BOTTOM,RIGHT,LEFT) + clip 2 line segments	2020, 2021, 2022, 2023, 2024	Compute codes via the 4 sign tests; AND both codes to classify Visible/Not-visible/Candidate; push the outside endpoint across whichever boundary its set bit indicates, iterate
Normalization transformation (window $\rightarrow$ viewport, both given)	2020, 2021, 2023(none), 2024	s_x, s_y from ratio of extents; $N =$ translate–scale–translate; watch for workstation-transform variant (NDC $\rightarrow$ device)
Normalization with aspect ratio preserved (only window given)	2022	Compute $a_w = \text{width/height}$; fill the longer NDC axis fully, scale the other axis by a_w so $s_x = s_y$
Sutherland–Hodgman polygon clipping (numeric or figure-based)	2021, 2022, 2023	Process one clip edge at a time; 4-case left/right test per subject edge; re-feed output to next edge
Weiler–Atherton polygon clipping (handles concave/multi-part output)	2024	Trace subject polygon; right-turn onto clip polygon at each exit; resume subject trace at next entry; output each closed loop
Convex/concave polygon identification + justification	2020, 2021, 2022	Check every interior angle $< 180^\circ$ (convex) or find one $\geq 180^\circ$ (concave, reflex vertex)

Exam-day strategy for Chapter 5: region-code clipping is pure mechanical book-keeping — compute codes, AND them, and always clip the *outside* endpoint first, one boundary at a time, recomputing the code after every intersection. For polygon clipping, default to Sutherland–Hodgman unless the question explicitly says Weiler–Atherton or the clip window is concave. The 2022 aspect-ratio-preserving normalization question is the one variant without an explicit viewport — don’t panic if only a

window is given, just apply the a_w rule from §1.1.